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A deep learning framework for environmental sound classification by fusing linear and nonlinear features

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Abstract

Environmental Sound Classification (ESC) is a crucial research direction in audio signal processing, aiming to identify and classify specific events in ambient sounds. Traditional methods typically rely on time–frequency features (e.g., inverse Mel spectrogram) fail to fully capture complex temporal patterns in sound signals. To address this critical issue, this paper employs Recurrence Plot (RP) to compensate for the deficiency of nonlinear features in time–frequency features. To validate the fused features and the RP of performance in ESC, comparative experiments were conducted on four different models (e.g., CNN and GRU) and two benchmark datasets. The model's performance based on deep learning is enhanced through integrating linear and nonlinear feature. Furthermore, a novel SResNet architecture is proposed, which embeds an attention mechanism into the feature fusion process of ResNet-18 and incorporates the SwiGLU activation function to optimize residual blocks. The smoothing property of SwiGLU contributes to stabilize residual networks and accelerate convergence, enabling the capture of more intricate patterns. Experimental results demonstrate that the proposed feature fusion outperforms traditional linear feature fusion methods on both ESC-50 and UrbanSound8K datasets, thereby validating the robustness of RP in ESC tasks. Concurrently, SResNet also exhibits superior performance compared to direct feature fusion. This innovative approach of parallel feature fusion and model optimization advances environmental sound analysis, enabling more comprehensive and accurate representation of ambient sound data.

Keywords: Sound event classification, Recurrence plot, Linear features, Feature fusion, SResNet

1 Introduction

ESC aims to assign semantic labels to audio clips, thereby facilitating automated the recognition of environmental sounds. Although less prominent than speech or music processing, ESC plays an indispensable role in specialized applications—including alarm systems for firefighting robots [1], road accident detection [2], and intelligent monitoring systems [3].

Conventional ESC methods predominantly rely on time–frequency features, such as those derived from short-time Fourier transform (STFT), Mel spectrograms, or

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wavelet scalograms. A key limitation of these representations, however, is their general disregard for phase information, which may result in an incomplete characterization of the acoustic signal's complexity [4, 5]. This shortcoming underscores the urgent need for advanced feature extraction techniques capable of capturing richer temporal dynamics. In this context, nonlinear dynamics offers a compelling alternative approach.

Phase space reconstruction techniques map time series into high-dimensional spaces, revealing intrinsic dynamical properties. Such methods have been successfully applied in diverse domains—from material defect detection [6–8] and musical timbre classification [9, 10] to EEG-based emotion recognition [11–13] and bioacoustic analysis [14, 15]. Central to this present work is the use of RP, which encode one-dimensional signals into two-dimensional images by capturing recurrent patterns and state transitions within phase space. This makes RP particularly suitable for characterizing nonlinear temporal structures in audio.

Meanwhile, recent advancements in CNN have demonstrated exceptional performance in image recognition and segmentation tasks [16–20]. Capitalizing on this, we propose a novel approach that bypasses traditional feature engineering: RP are used directly as input to deep learning architectures, allowing the model to automatically learn discriminative representations automatically from recurrence visualizations.

Notably, our approach integrates RP with conventional time–frequency analysis within a complementary, dual-perspective framework. While spectrograms offer a linear time–frequency decomposition—highlighting harmonic structures and energy distributions over time—RP reveal the underlying nonlinear dynamics and deterministic chaotic behavior of the signal. Together, they provide a multi-level characterization: RP probe microscopic dynamical properties, while spectrograms describe macroscopic spectral patterns. We also explore threshold-free RP variants inspired by Dong et al. [21]

This paper further proposes an enhanced ResNet-18 model, designated as SResNet. The model incorporates a dual-modal feature fusion (DMFF) module [22] subsequent to the input stage, enabling the structured deep integration of input features. It supports features extracted by various backbone networks, including ResNet and VGG, thereby demonstrating strong adaptability and scalability.

Meanwhile, SwiGLU gated residual block in SResNet integrates with ResNet's residual blocks to control information flow through gating mechanisms, thereby enhancing CNN capability in processing sequential data (e.g., text). The SwiGLU architecture proposed in this paper is based on Dauphin et al.'s [23] method and incorporates Shazeer's [24] research, employing the Swish function as the gating activation function to form a SwiGLU variant that optimizes gradient flow.

In conclusion, this paper proposes a feature fusion framework, which combines Mel-spectrogram-based and RP-based pathways. The first pathway extracts high-resolution time–frequency representations; the second pathway employs phase space reconstruction to generate RP, which are then processed using dilated convolutions to capture multi-scale topological features. Finally, the effectiveness of the extracted RP and the proposed framework in ESC is verified through multiple deep learning models, and comparative analyses are conducted with the proposed SResNet.

2 Related theories

This section elaborates on the techniques employed to extract RP from audio signals. A primary challenge in directly analyzing audio time series is their considerable length, which often encompasses hundreds of thousands of points. To address the computational challenges associated with such long sequences and to facilitate effective nonlinear feature extraction via RP, we apply a dimensionality reduction pipeline consisting of endpoint detection (EPD) and piecewise aggregate approximation (PAA). The EPD step is implemented using the short-time entropy spectrum (STES) method, the purpose of STES is to cut the longest time series of valid audio in the sound event.

2.1 STES

The core idea behind the STES method is that the spectral energy of speech signals is more concentrated than that of noise, resulting in lower spectral entropy. In contrast, the spectral energy of noise is typically more uniformly distributed, leading to a higher entropy value. By computing the short-time spectral entropy for each frame, speech segments can be effectively distinguished from non-speech segments.

The procedure for computing the STES is as follows:

- (1) Framing and Windowing: The input signal $x(n)$ is first divided into short, overlapping frames (typically 20–30 ms in length with a 10 ms frame shift). Each frame is then multiplied by a window function (e.g., a Hamming window).
- (2) Power Spectrum Computation: For each frame, the fast Fourier transform (FFT) is applied to obtain the frequency spectrum $X(k)$. The power spectrum is computed as follow.

$$P(k) = |X(k)|^2 \quad (1)$$

- (3) Probability Density Estimation: The power spectrum is normalized to form a probability density function over the positive frequency bins (where N is the FFT length):

$$p(k) = \frac{P(k)}{\sum_{k=1}^N P(k)} \quad (2)$$

- (4) Short-Time Spectral Entropy Calculation:

$$H = - \sum_{k=1}^N p(k) \log_2 p(k) \quad (3)$$

- (5) Thresholding and Decision: A threshold T is defined. A frame is classified as a speech frame if its entropy H is less than T ; otherwise, it is classified as a non-speech frame.

2.2 PAA

As a classical technique for time series dimensionality reduction and feature representation, PAA is widely employed in domains including time series data mining, pattern recognition, and signal processing. The core principle of PAA is to segment the original time series into a number of contiguous, equal-length intervals and to represent each interval by its mean value. This process achieves significant data reduction while preserving the essential trend information of the original sequence.

Given a time series $X = \{x_1, x_2, \dots, x_N\}$, the PAA representation reduces its dimensionality by dividing it into M segments of equal length. (When N is not evenly divisible by M , common techniques such as sliding window or zero-padding can be applied to handle the remainder.) The mean value for each segment is calculated as follows:

$$\bar{x}_i = \frac{M}{N} \sum_{j=\frac{N}{M}(i-1)+1}^{\frac{N}{M}i} x_j, \quad i = 1, 2, 3, \dots, M \quad (4)$$

Finally, PAA outputs a dimensionality-reduced sequence, $\bar{X} = \{\bar{x}_1, \bar{x}_2, \dots, \bar{x}_M\}$ where $M \ll N$.

2.3 RP

RP is a visualization technique introduced by Eckmann et al. in 1987. It is designed to quantify the recurrences of states in a time series. The method constructs a two-dimensional matrix that reveals the underlying dynamics of the system, such as periodicity, chaotic behavior, and state transitions, providing a visual representation of the characteristics of nonlinear time series.

The RP methodology is grounded in phase space reconstruction theory, which aims to describe the recurrence properties of trajectories in the reconstructed state space. The formal procedure for constructing an RP is defined as follows:

- (1) Given a time series $\{x_i\}_{i=1}^N$, the state vector is obtained first by phase space reconstruction (usually using time delay embedding method):

$$y_i = (x_i, x_{i+\tau}, \dots, x_{i+(m-1)\tau}) \quad (5)$$

where m is the embedding dimension, τ is the time delay.

- (2) Calculating the distance between point i and point j in the reconstructed phase space:

$$D_{i,j} = \|y_i - y_j\|, \quad i, j = 1, 2, 3, \dots, N - (m-1)\tau \quad (6)$$

$\|\cdot\|$ is usually the Euclidean norm.

- (3) The recurrence matrix $R_{i,j}$ is defined as:

$$R_{i,j} = H(\varepsilon - D_{i,j}), \quad i, j = 1, \dots, N \quad (7)$$

where $H(x)$ is the Heaviside step function and ε is the distance threshold.

$$\begin{cases} H(x) = 1, & x > 0 \\ H(x) = 0, & x < 0 \end{cases} \quad (8)$$

Therefore, the recurrence matrix is a square array composed of 0 and 1. When

$$D_{i,j} > \varepsilon, R_{i,j} = 0; \quad \text{when } D_{i,j} < \varepsilon, R_{i,j} = 1.$$

(4) Visualization

The RP is visualized as a two-dimensional image by plotting the indices i of the time series on the horizontal axis and j on the vertical axis (Fig. 1). In this image, a recurrence $R_{i,j}=1$ is typically represented by a black dot, while the absence of a recurrence $R_{i,j}=0$ is represented by a white dot. By definition, $R_{i,j}=1$ holds true for all positions, ensuring the network always maintains a dominant diagonal (LOI). Furthermore, since $R_{i,j}=R_{j,i}$, all RP exhibit symmetry about this diagonal. Figure 1 shows RP of a sheep bleating with a threshold of 0.1. The time series has a length of 30,000, with black dots representing 1 and white dots representing 0.

3 Proposed method

This paper proposes a method for ESC that leverages RP to comprehensively characterize nonlinear temporal features in acoustic signals, which are then integrated with standard spectral features. The deep learning models utilized in our experiments include VGG-16, ResNet-18, a CNN-GRU hybrid, and a SResNet architecture. Notably, the SResNet architecture is enhanced by incorporating a specific SwiGLU module to suppress redundant information and extract more discriminative features.

3.1 Nonlinear feature extraction methods

A prevalent approach for handling time series involves transforming the one-dimensional sequence into a two-dimensional image representation for subsequent deep learning

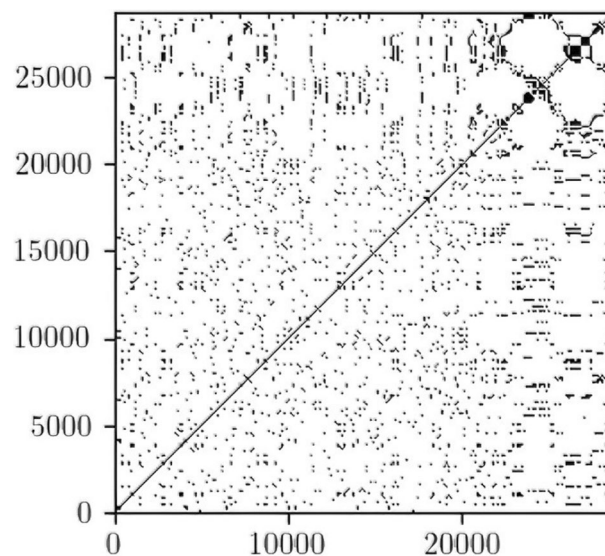


Fig. 1 An instance of a Recurrence plot generated from a sheep bleating sound signal, with a distance threshold set at 0.1

analysis. Our method follows this paradigm by generating RP for input to deep learning models.

The procedure begins with normalizing the sound event's time series and applying end-point detection (EPD) to isolate the longest active segment. PAA is then employed to reduce the sequence length by half, primarily to alleviate computational burden. Nonlinear feature extraction is initiated by generating the RP from this shortened time series. Given the significant dimensional disparity between the extracted RP image and the spectral features (e.g., Mel-spectrograms), a dilation convolution operation is applied to compress the RP features. Specifically, the large-scale spectrogram is scaled to three times its original spectral dimension. Subsequently, for effective fusion with the linear features, the dimensions of the features from the two parallel streams (RP and spectrogram) are adjusted accordingly before being merged. The complete nonlinear feature extraction pipeline is illustrated in Fig. 2.

3.2 Linear acoustic features

A set of representative linear features was selected for audio signal analysis in this study. Specifically, the adopted features comprise the logarithmic Mel spectrogram (Logmel), gammatone Frequency Cepstral coefficients (GFCC), Autocorrelation coefficients (AC), the constant-Q transform (CQT), the standard Mel-frequency cepstral coefficients (MFCC), and their first-order derivatives (Δ MFCC). These features capture the characteristics of audio signals from diverse perspectives, collectively providing a comprehensive set of linear feature representations for the subsequent classification task.

3.3 DMFF

The process begins with spatial feature compression using SFS to generate Tokens for each modal feature, which are then encoded with positional information as input to the CFE module. To address modal misalignment, dual CFE modules are employed to enhance complementary feature learning, with each module operating independently without shared weights. Figure 3 illustrates the architecture of the dual-modal feature fusion.

3.4 Introduction of models

The VGG-16 architecture comprises 16 learnable weight layers, specifically consisting of 13 convolutional layers and three fully connected layers. Its core characteristic lies in uniformly adopting 3×3 smallest convolution kernels and 2×2 maximum pooling windows. By strictly maintaining a convolution stride of 1 and a padding value of 1, it ensures the spatial resolution of the feature maps is completely preserved during the convolution process. The front end of the network adopts two consecutive 3×3 convolutional layers as the basic module, while the back end extends to a deep structure with three consecutive convolutional layers. This progressively deepening strategy not only effectively expands the receptive field (equivalent to the perception range of 5×5 and 7×7 convolution kernels), but also significantly alleviates the vanishing



Fig. 2 Nonlinear feature extraction flow chart, dilated convolution (Rate = 2) for Feature downscaling

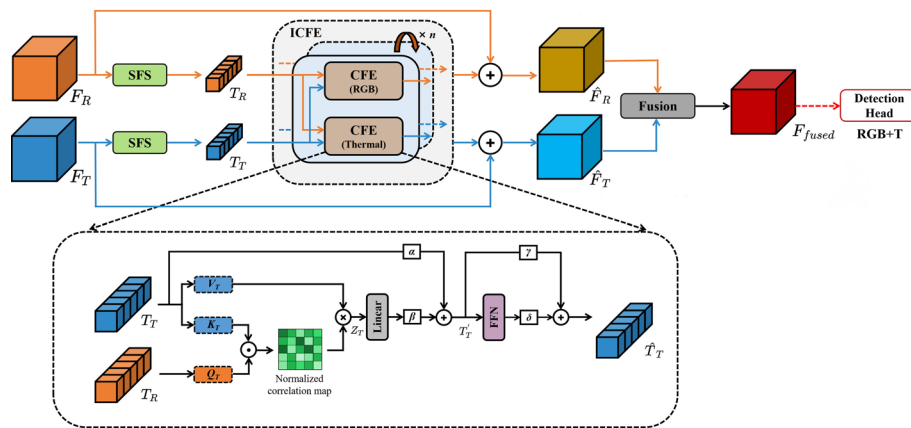


Fig. 3 Illustration of the proposed DMFF module

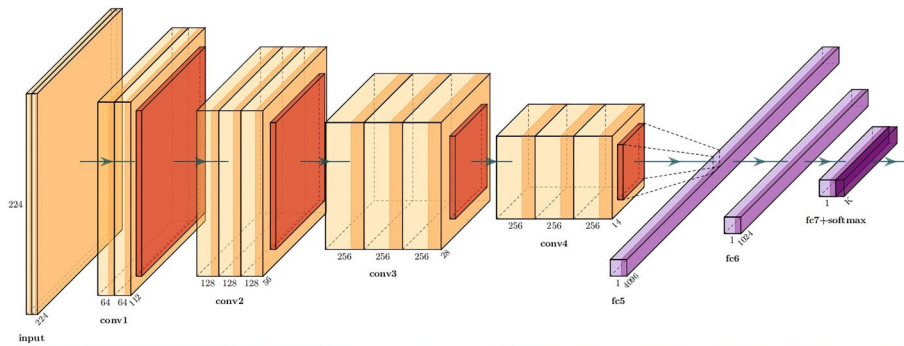


Fig. 4 Architecture of VGG-16

gradient problem in deep networks by introducing ReLU activation functions and batch normalization technology. In terms of feature dimensions, VGG-16 follows an exponential growth pattern from 64 to 256. Spatial downsampling and channel dimension enhancement are achieved through maximum pooling layers, and finally, high-level semantic feature abstraction and classification decision are completed through a three-layer fully connected network ($4096 \rightarrow 1024 \rightarrow 50$). Figure 4 shows the architecture of VGG-16, with input being two-dimensional data (224×224) formed by the fusion of linear features and nonlinear features.

CNN-GRU constructs a cascaded feature extraction architecture, aiming to simultaneously capture spatiotemporal correlations in the data. Through a parallel submodule design, the system achieves the decoupling and cooperative optimization of multidimensional feature learning. Figure 5 shows the architecture of CNN-GRU, which is composed of VGG-16 and GRU connected in parallel.

The GRU submodule, as the core of temporal feature extraction, adopts a simplified gating mechanism to balance performance and computational efficiency, with its detailed architecture illustrated in Fig. 6. While maintaining the essence of the LSTM gating mechanism, the GRU innovatively integrates the forget gate and the input gate into a unified update gate, and introduces a reset gate to achieve adaptive

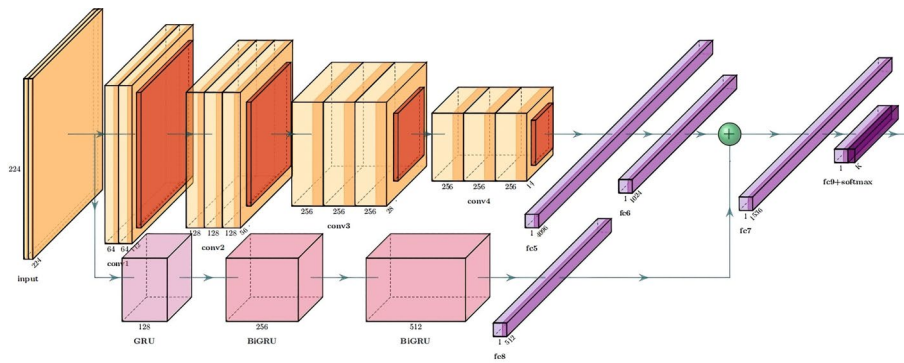


Fig. 5 Architecture of CNN-GRU

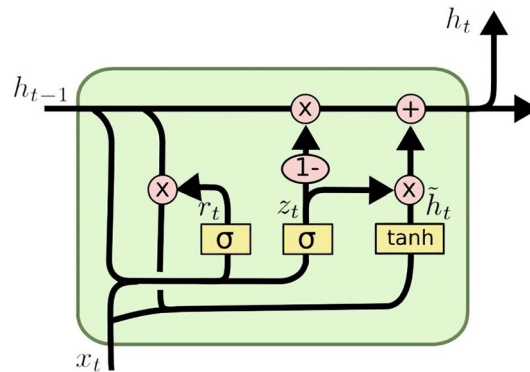


Fig. 6 Architecture of GRU

filtering of historical information, thereby reducing the parameter scale to 75% of LSTM and significantly improving training efficiency. Its forward propagation process can be formalized as follows: the update gate $z_t = \sigma(Wz \cdot [h_{t-1}, x_t])$ controls the retention ratio of the hidden state from the previous time step, the reset gate $r_t = \sigma(Wr \cdot [h_{t-1}, x_t])$ regulates the degree of fusion between the current input and historical information, the candidate hidden state $\tilde{h}_t = \tanh(W[\tilde{r}_t \odot h_{t-1}, x_t])$ generates new memory content through a nonlinear transformation, and the final hidden state $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$ achieves a smooth transition of information. Figure 6 shows the architecture of GRU.

ResNet-18 architecture contains 17 weight layers and one fully connected layer, consisting of four residual block modules. Each module stacks two basic blocks, adopts a “ $1 \times 1-3 \times 3-1 \times 1$ ” convolution structure, and applies batch normalization and ReLU activation function at the end of each convolutional path to ensure gradient stability. Shortcut connections achieve identity mapping through element-wise addition, while 1×1 convolution performs linear projection when dimensions do not match to maintain feature consistency. The network uses 3×3 large convolution kernels combined with max pooling for fast downscaling, followed by residual blocks with a stride of 2 to gradually compress the spatial dimensions. The number of channels expands exponentially from 32 to 512. When creating shortcut connections, the system applies a downsampling layer to reduce sampling and dimensions. Finally, high-level semantic

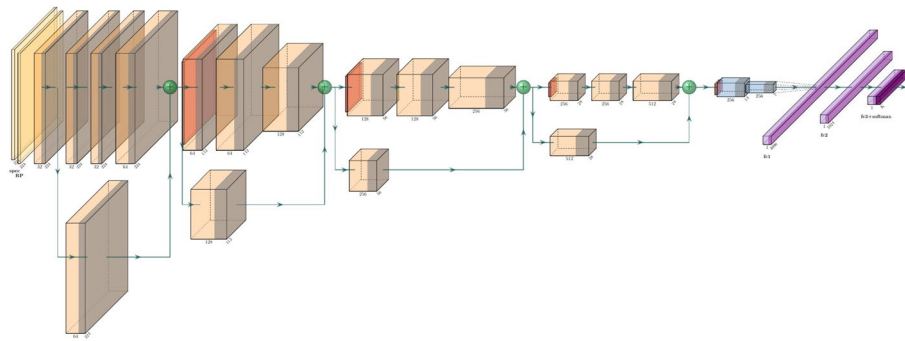


Fig. 7 Architecture of ResNet-18, consisting of four residual blocks

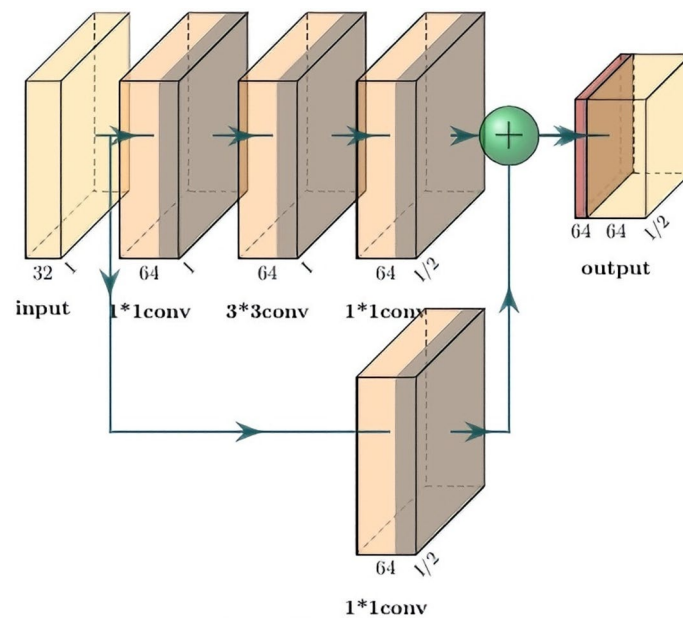


Fig. 8 Architecture of residual blocks

feature abstraction and classification decision are achieved through a three-layer fully connected network ($4096 \rightarrow 1024 \rightarrow 50$). Figure 7 shows the ResNet-18 architecture composed of four serially connected residual blocks.

The residual block, as the core building module of ResNet, has two unique paths: the main path contains multiple convolutional layers for learning residuals, and a shortcut connection that directly passes the input to the output of the main path. This design ensures information can be seamlessly transmitted to subsequent layers, promotes gradient propagation, and alleviates the vanishing gradient problem. The implementation scheme proposed in this paper adopts the first path, namely $H(x) = F(x) + x$. Figure 8 shows the architecture of the residual block.

The SResNet model builds upon the classic ResNet-18 architecture with two key innovations: (1) The introduction of the SwiGLU gating mechanism in basic residual blocks to enhance nonlinear feature extraction; (2) The implementation of a DMFF module at the network's start, inspired by ICAFusion, to achieve deep integration of local and

global features. The core innovation of this module is the use of cross-attention mechanism to guide the fusion of dual-path features, rather than simple concatenation or addition. The overall architecture of SResNet is illustrated in Fig. 9.

The core of the SwiGLU gated residual block is the SwiGLU operation, defined as $\text{Output} = (\text{Linear}(X)) \otimes \text{Swish}(\text{Linear}(X))$. Layer 1: A 1×1 convolution to extract spatial features, followed by batch normalization (BN) and Swish activation function. The output channels here are expanded to $2k$. This splits the feature map into two equal parts (each with k channels) along the channel dimension for the SwiGLU unit. As shown in Fig. 10, one part generates a gating signal via the Swish function, while the other serves as the main feature. The two are then multiplied to complete the gating process. Projection layer: The SwiGLU output channel count is k . To match the shortcut connection, a 1×1 convolution is sometimes added to adjust the channel count. Figure 10 shows the SwiGLU-gated residual block.

4 Experiments and results

4.1 Datasets

The ESC-50 dataset contains 2000 environmental audio samples, divided into five major categories: animal sounds, natural soundscapes and water sounds, human non-speech sounds, interior or domestic sounds, and exterior or urban noises. Each category is further subdivided into ten subcategories. To improve data quality, the data augmentation techniques used in this experiment process each audio sample: the duration of a single sample is extended to 10 s through repeated sampling and optimized using time stretching and pitch shifting techniques. After these processes, the original 2000 samples are increased to 6000. The UrbanSound8K dataset contains 8732 urban environmental audio clips with a duration not exceeding 4 s, covering 10 typical scenarios including air conditioner, car horn, children playing, dog bark, drilling, engine idling, gun shot, jackhammer, siren, and street music. The audio files are classified into ten folders from fold1 to fold10, with each folder containing approximately 873 wav format audio files. The experimental benchmark tests use two common feature sets, ASR and ESC, while the input features for the neural network models come from the processed spectrograms and RP features.

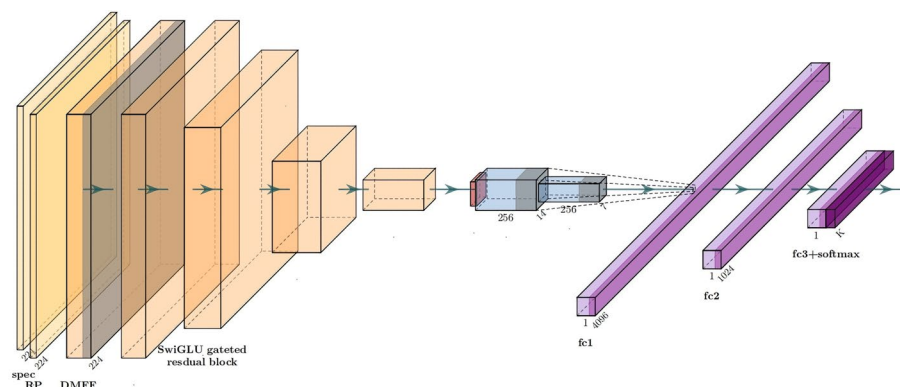


Fig. 9 Architecture of SResNet, including a feature fusion module (DMFF) and four SwiGLU gated residual blocks

Table 1 Compares the classification accuracy of various features and their fusions on the UrbanSound8K dataset based on the ResNet-18 model

Nonlinear features	Linear features	Accuracy (%)
–	Logmel	91.40
Threshold-free RP	Logmel	94.34
RP with a threshold of 0.2	Logmel	91.86
–	Logmel + GFCC + AC	97.67
Thresholded-free RP	Logmel + GFCC + AC	97.99
–	Logmel + GFCC	97.15
Thresholded-free RP	Logmel + GFCC	97.84
Thresholded-free RP	–	86.28

Table 2 Classification accuracy of feature fusion based on VGG16. The table compares the classification accuracy of linear features (e.g., Logmel, MFCC) and their fusion with unthresholded RP on ESC-50 and UrbanSound8K datasets. Results verify the complementary advantages of linear and nonlinear features

Nonlinear features	Linear features	Accuracy	
		ESC-50 (%)	UrbanSound8K (%)
–	Logmel + MFCC + Δ MFCC	95.95	98.07
Thresholded-free RP	Logmel + MFCC + Δ MFCC	96.53	98.37
–	Logmel	93.75	94.37
Thresholded-free RP	Logmel	95.76	96.11
–	MFCC	95.76	95.41
Thresholded-free RP	MFCC	96.04	96.30
–	Logmel + MFCC	95.94	96.13
Thresholded-free RP	Logmel + MFCC	96.06	97.19
Thresholded-free RP	–	94.68	95.63

(1) Experiments were conducted on the UrbanSound8K dataset based on the ResNet-18.

As shown in Table 1, when using RP alone combined with the ResNet-18 network, the accuracy reached 86.28%. After fusing RP with Logmel, the accuracy increased by 2.94%. When RP was fused with multiple spectrograms, the accuracy further improved by 0.32%. It is worth noting that the RP without threshold processing performed better in terms of recognition accuracy, achieving a higher accuracy rate. In subsequent experiments, we use the RP without threshold processing as the nonlinear feature for analysis.

(2) Experiments were conducted on the ESC-50 and UrbanSound8K datasets based on the VGG16.

As presented in the Table 2, RP without threshold processing achieves promising classification accuracy, and its fusion with linear features further enhances performance. A comprehensive analysis of the table reveals two key insights: first, recognition accuracy tends to increase with the number of integrated features; second, Thresholded-free RP contributes to higher accuracy compared to linear features alone. Furthermore, the fusion of linear and nonlinear features consistently achieves the best performance (e.g.,

Table 3 Feature fusion results on ESC-50 via CNN-GRU. Recognition accuracy of different linear feature combinations (with/without unthresholded RP) evaluated on the ESC-50 dataset using the CNN-GRU hybrid model. More fused features generally improve accuracy

Nonlinear features	Linear features	Accuracy (%)
–	Logco + Logmel + LogCQT	96.10
Thresholded-free RP	Logco + Logmel + LogCQT	96.21
–	Logmel	95.35
Thresholded-free RP	Logmel	95.63
Thresholded-free RP	–	93.36

Table 4 Feature fusion performance of SResNet. Classification accuracy of linear features (Logmel, GFCC, AC) and their fusion with unthresholded RP on ESC-50 and UrbanSound8K datasets. The SResNet model demonstrates superior robustness in multi-feature fusion

Nonlinear features	Linear features	Accuracy	
		ESC-50 (%)	UrbanSound8K (%)
Thresholded-free RP	Logmel + GFCC + AC	97.59	97.34
–	Logmel + GFCC + AC	96.67	96.49
Thresholded-free RP	Logmel + GFCC	97.84	98.39
–	Logmel + GFCC	97.15	97.26
–	AC	88.16	94.68
Thresholded-free RP	AC	93.75	97.17
–	GFCC	97.50	96.44
Thresholded-free RP	GFCC	97.99	97.89
Thresholded-free RP	–	97.82	95.46

96.53% on ESC-50 and 98.37% on UrbanSound8K for the combination of Thresholded-free RP with Logmel + MFCC + Δ MFCC), which verifies the complementary advantages of linear and nonlinear feature representations.

(3) Experiments were conducted on the ESC-50 dataset based on the CNN-GRU.

As shown in Table 3, the RP accuracy of the CNN-GRU also reached 93.36%, which is 2% higher than the linear feature fusion method. Furthermore, the more features are fused, the higher the accuracy. Recognition accuracy of different linear feature combinations (with/without unthresholded RP) evaluated on the ESC-50 dataset using the CNN-GRU hybrid model. More fused features generally improve accuracy.

(4) Experiments were conducted on the ESC-50 and UrbanSound8K dataset based on the SResNet.

Experimental results in Table 4 show that as a nonlinear feature, threshold-free RP demonstrates stable enhancement of recognition accuracy for both linear and fusion features, with the most significant improvement observed in the less effective AC feature (5.59 percentage points on the ESC-50 dataset), validating its complementary

enhancement value to linear features. The optimal feature configuration is clearly defined: “Threshold-free RP + Logmel + GFCC” achieves the highest accuracy of 98.39% on the UrbanSound8K dataset, while “Threshold-free RP + GFCC” serves as a sub-optimal choice that balances accuracy (97.99% on ESC-50) with simplicity. Multi-feature fusion does not imply that more features are always better. The core strategy for improving model performance lies in selecting high-quality features (e.g., Logmel and GFCC) and combining them with complementary nonlinear features, while avoiding redundant features like AC to minimize performance degradation. The analysis further reveals that the SResNet model demonstrates superior robustness in multi-feature fusion scenarios. Quantitative comparisons across four experimental datasets indicate that SResNet achieves optimal accuracy in recognition tasks utilizing RP features without thresholding. These results conclusively demonstrate the model’s significant advantages in feature adaptation, fusion efficiency, and classification performance.

(6) Comparing the accuracy of Thresholded-free RP in different models.

As shown in Fig. 11, by comparing the performance of the four models on the ESC-50 and UrbanSound8K dataset, it can be clearly seen that SResNet performs the best, followed by VGG-16 which is commonly used for image recognition, while ResNet-18 has the worst performance. This comparison result also provides new ideas for future research: using multiple models to identify different features, and then performing feature fusion through flattening operations. Moreover, the simple feature concatenation method yields lower accuracy than the SResNet model with feature fusion.

(6) Comparison of SResNet with existing models.

As shown in Table 5, SResNet achieves recognition accuracy of 98.39% on the UrbanSound8K dataset and 97.84% on the ESC-50 dataset, outperforming all baseline models. On ESC-50, SResNet surpasses the second-best model S-HQCNN (93.75%) by 3.59 percentage points, the classic hybrid architecture CNN-RNN (91.50%) by 5.84 percentage points, and the traditional deep learning model GoogleLeNet (73.00%) by 24.34 percentage points. On US8K, SResNet outperforms CNN-RNN (98.00%) by 0.39 percentage points and all other comparison models, demonstrating exceptional

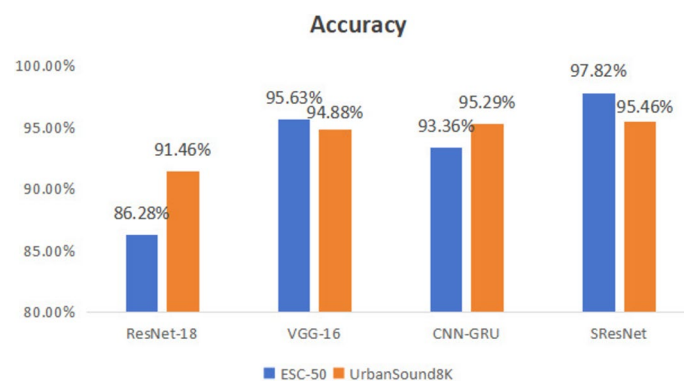


Fig. 11 Accuracy of thresholded-free RP in different models on ESC-50 and UrbanSound8K dataset

classification performance. By introducing an attention mechanism to achieve adaptive multi-feature fusion, SResNet comprehensively outperforms baseline models without attention enhancement or traditional feature fusion strategies, even surpassing the complex hybrid architecture CNN-RNN. These results conclusively prove that SResNet's feature fusion mechanism more efficiently extracts complementary information between features, enhances model adaptability to complex acoustic scenarios, and highlights the scientific rigor and superiority in its architectural design.

5 Conclusion

This study focuses on ESC, a critical research direction in audio signal processing. To address the limitation of traditional methods that rely on time–frequency features and thus fail to fully capture the complex temporal patterns of sound signals, we propose a deep learning framework integrating linear and nonlinear features, and design an optimized model SResNet to achieve more accurate and comprehensive environmental sound recognition. Meanwhile, the SResNet model integrating a dual-modal feature fusion (DMFF) module with the SwiGLU activation function, forming a dual-path fusion technical solution combining “linear features + nonlinear features”. The innovation and value of this paper are primarily reflected in three aspects: First, it is the first to apply threshold-free RP to the ESC task, demonstrating its effectiveness and robustness in capturing nonlinear temporal patterns. Second, the proposed SResNet model, through the optimization of the DMFF module and SwiGLU activation function, provides a new architectural design approach for deep feature fusion and model performance enhancement. Third, it establishes a dual-path fusion framework combining “linear features + nonlinear features”, breaking through the limitations of traditional single-feature dependency and offering a more comprehensive feature representation scheme for environmental sound analysis. This research can be widely applied to practical scenarios such as fire robot alarms, road accident detection, and intelligent monitoring, providing technical support for precise acoustic signal recognition.

Table 5 Comparison of SResNet with existing models

Model	Accuracy	
	US8K (%)	ESC-50 (%)
GoogleLeNet [25]	93.00	73.00
CNN [26]	78.14	81.40
FTAP [27]	–	88.20
CNN-RNN [28]	98.00	91.50
EnVITSNA [29]	83.20	85.85
TD-CML [30]	–	85.75
AM-ESC [31]	93.77	–
CNN-Att [32]	95.00	–
SQNN [33]	87.51	90.00
St-HQCNN [34]	97.14	93.75
SResNet(ours)	98.39	97.84

Author contributions

Canhui Zhou was contributed writing—original draft, visualization, project administration, methodology, and conceptualization. Jinfang Zeng was involved in writing—original draft, visualization, validation, and methodology. Dian Cai was performed validation and methodology.

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Data availability

Data will be made available on request.

Declarations**Declaration of Generative Artificial Intelligence and AI-Assisted Technologies in Manuscript Preparation**

No generative artificial intelligence tools or AI-assisted technologies were used in the preparation of this manuscript. All content, including text, figures, and tables, was created and edited by the authors.

Competing interests

The authors declare no competing interests.

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